

MANN KHADELWAL

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EDUCATION

Vellore Institute of Technology — Bhopal, Madhya Pradesh

B.Tech in Electronics and Communication Engineering (Expected October 2027)

- Cumulative GPA: 7.61/10

Frame International School — Dausa, Rajasthan

12th Standard (CBSE) (July 2023)

- Percentage: 76%

Frame International School — Dausa, Rajasthan

10th Standard (CBSE) (July 2021)

- Percentage: 92%

EXPERIENCES

Team Captain | Team VITAKSH (May 2025 - April 2026)

Captain of the student satellite team of VIT Bhopal.

- Led a **multidisciplinary team of 25+ members** in designing and launching a high-altitude BalloonSat.
- Oversaw **end-to-end system integration**: payload, telemetry, communication, and ground station.

Intern in Research Center Imarat DRDO | Hyderabad (April 2025 – July 2025)

- Designed microstrip patch antennas for the frequencies (2.25 GHz, 2.8 GHz, 1.575 GHz).
- Worked on antenna design using HFSS and MATLAB.
- Tuned fabricated microstrip patch antennas through iterative trimming and simulation comparison.

PROJECTS

Ultrasonic Array Radar System (Jan 2025 - Apr 2025)

- Using an array of ultrasonic sensors for object detection and distance measurement.
- Locating the exact location in terms of azimuth angle, elevation angle, and distance.
- Mapping the object into a heatmap image.

5.5 GHz Bandpass Filter Design (Dec 2025 - Jan 2026)

- Using an array of ultrasonic sensors for object detection and distance measurement.
- Locating the exact location in terms of azimuth angle, elevation angle, and distance.
- Mapping the object into a heatmap image.

Slotted Flower-Shaped Coaxial Feed Antenna (Radar Application)

- Designed an innovative flower-shaped slotted antenna with coaxial feed for improved radiation characteristics.
- Analyzed S11, gain, and radiation patterns using HFSS.
- Currently extending the design into an antenna array for radar applications to enhance directivity and gain.

1×4 Microstrip Patch Array for 24 GHz Radar

- Designed a 1×4 patch antenna array at 24 GHz for radar applications.
- Developed a corporate feed network with power divider for equal power distribution.
- Optimized array performance for high gain and beam steering potential.

SIW Horn Antenna Design

- Designed a Substrate Integrated Waveguide (SIW) horn antenna for high-frequency applications.
- Combined advantages of waveguide structures and planar circuits for low loss and compact design.
- Analyzed S-parameters (S11), gain, and radiation patterns using HFSS.

Non-Invasive Glucose Monitoring System (Jul 2025 - Present)

- Designing an unalike microstrip patch antenna radiating at 2.45 GHz.
- Determining the dielectric properties of blood with different glucose concentrations.
- Engineered a machine learning model to predict glucose concentration based on dielectric properties, integrated into a web application.

High-Altitude Balloon Satellite (BalloonSat) | Team VITAKSH | (Aug 2025 - March2026)

- Designed and developed a BalloonSat system for high-altitude data collection and communication.
- Establishing the telemetry of 25 - 30 Km of range with an ordinary transreceiver module.
- Sensors integration and circuit design.
- Designing an antenna for the telemetry.

SKILLS

- Front-end Development
- Circuit Designing
- MATLAB
- Antenna Design and Simulation
- High Frequency Structure Simulator
- Filter Designing
- Communication Skills